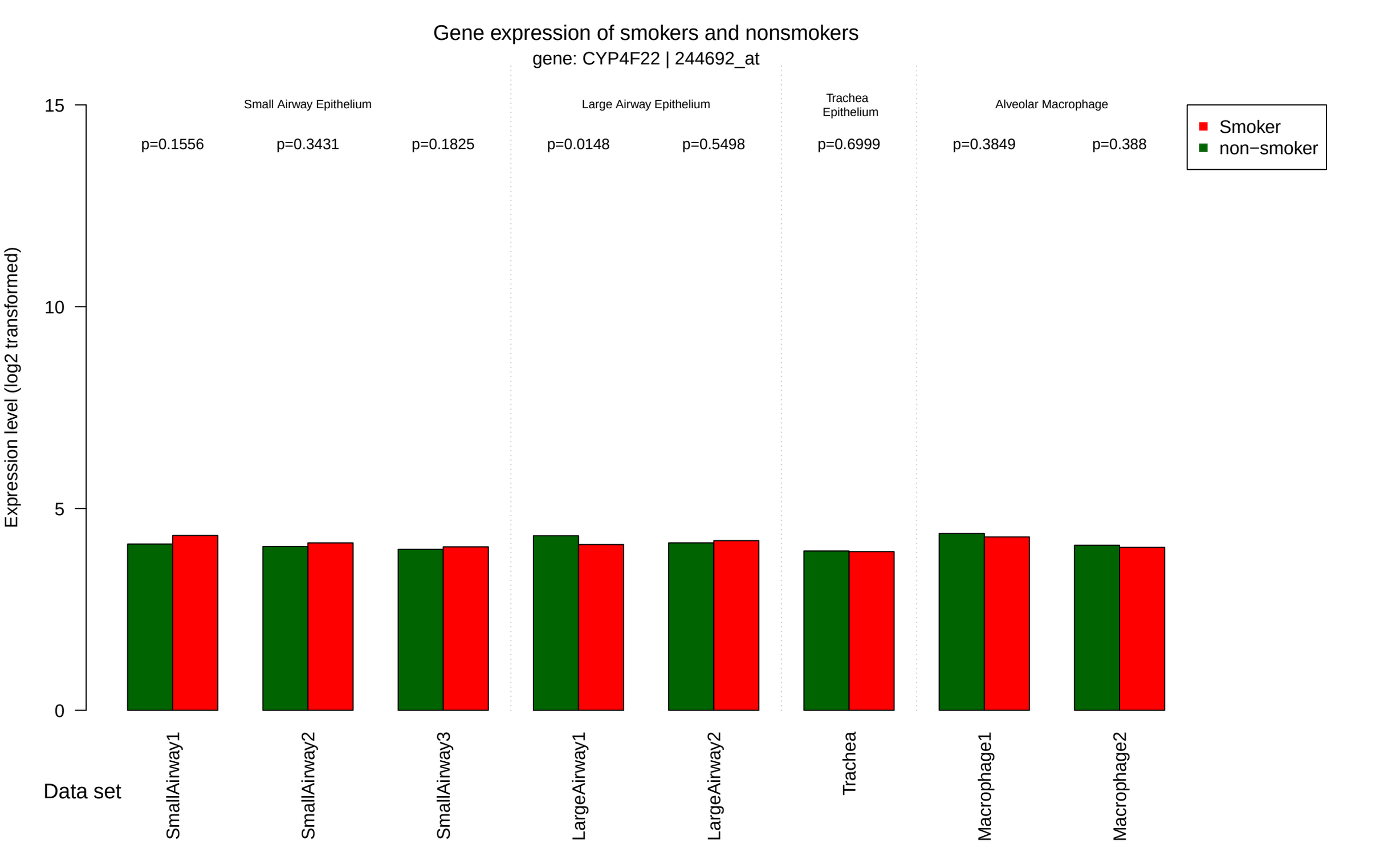
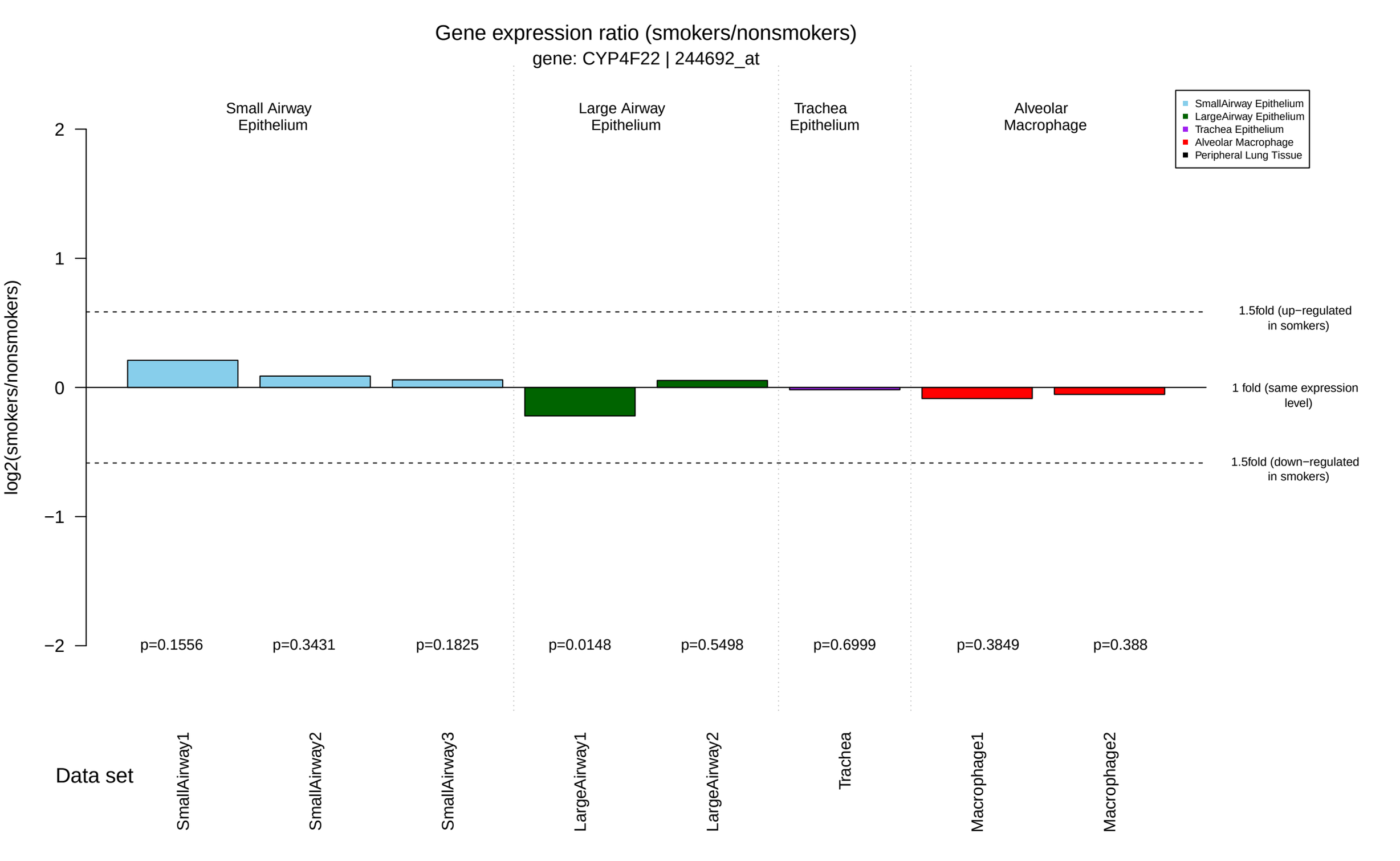
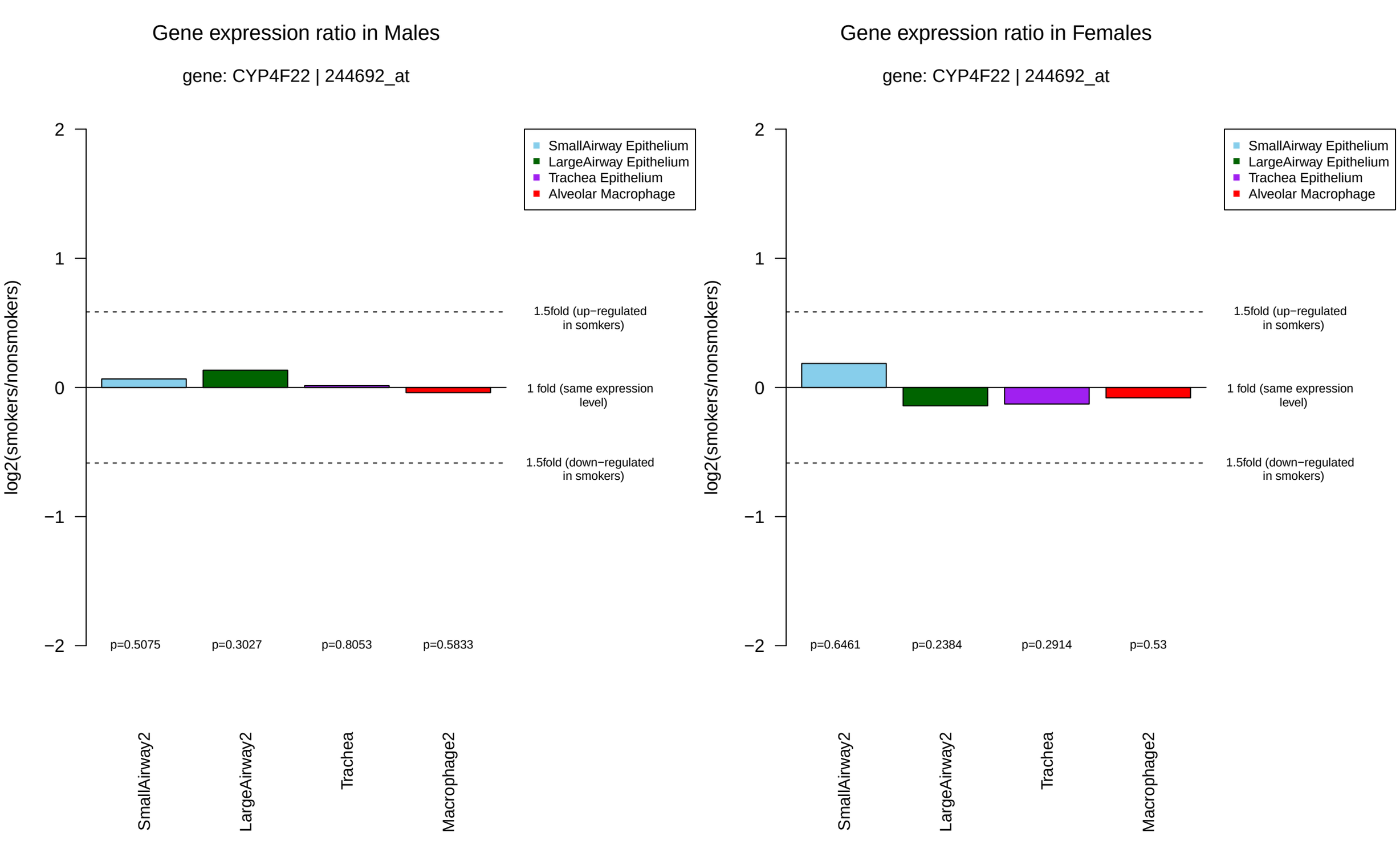
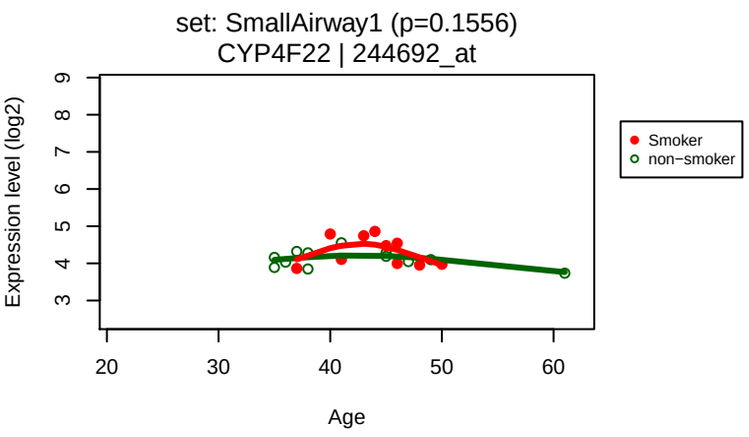
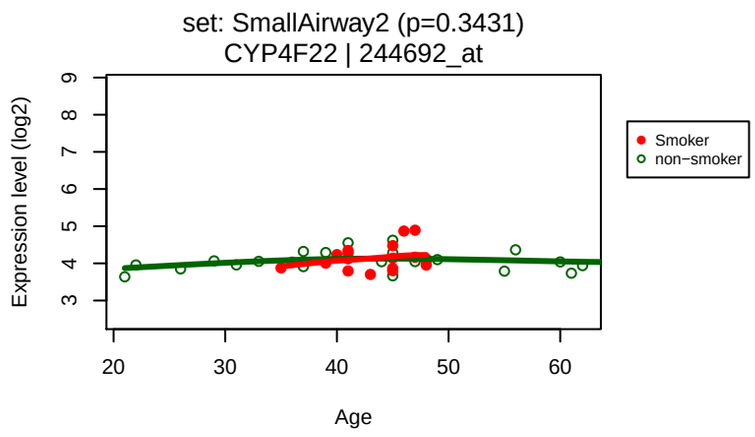
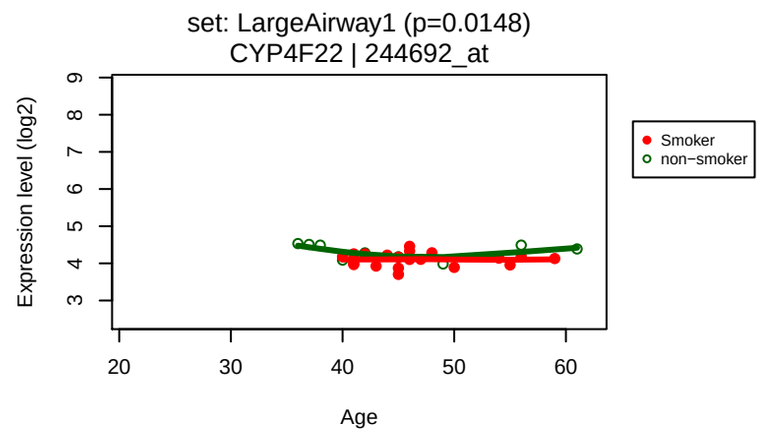

Gene expression of smokers and nonsmokers
gene: CYP4F22 | 244692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1556
p=0.3431
p=0.1825
p=0.0148
p=0.5498
p=0.6999
p=0.3849
p=0.388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

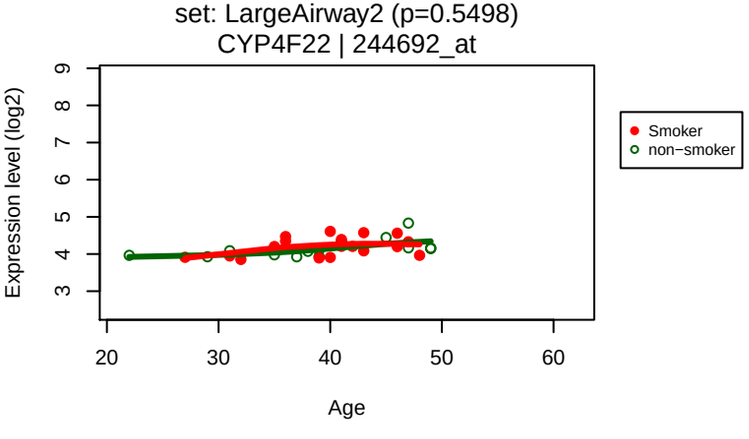

Gene expression ratio (smokers/nonsmokers)
gene: CYP4F22 | 244692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1556
p=0.3431
p=0.1825
p=0.0148
p=0.5498
p=0.6999
p=0.3849
p=0.388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

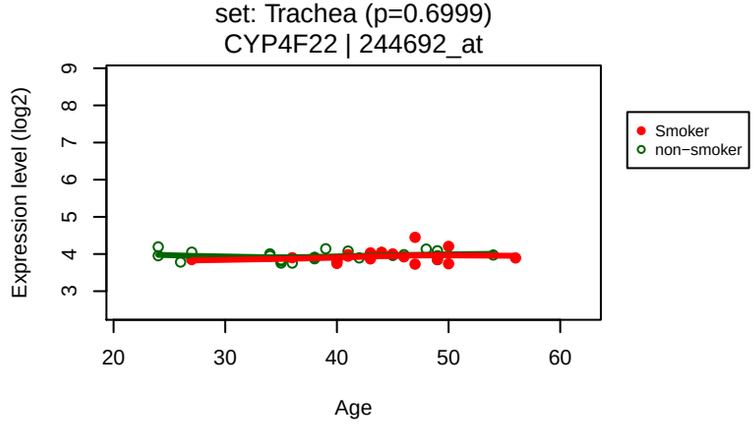

Gene expression ratio in Males
gene: CYP4F22 | 244692_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5075
p=0.3027
p=0.8053
p=0.5833
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP4F22 | 244692_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6461
p=0.2384
p=0.2914
p=0.53
SmallAirway2
LargeAirway2
Trachea
Macrophage2

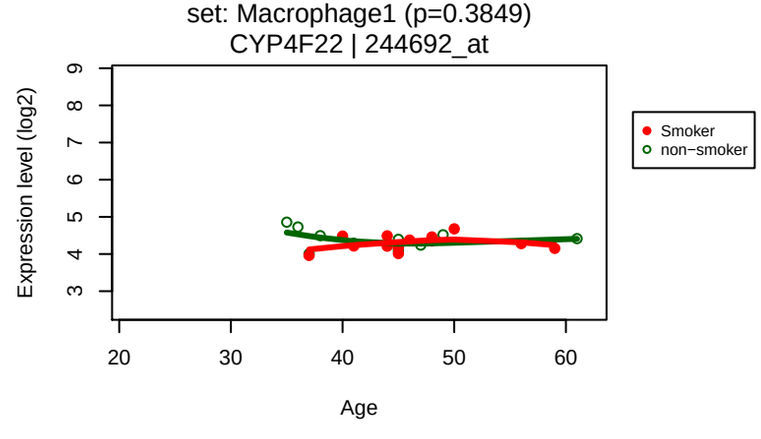

set: SmallAirway1 (p=0.1556)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker

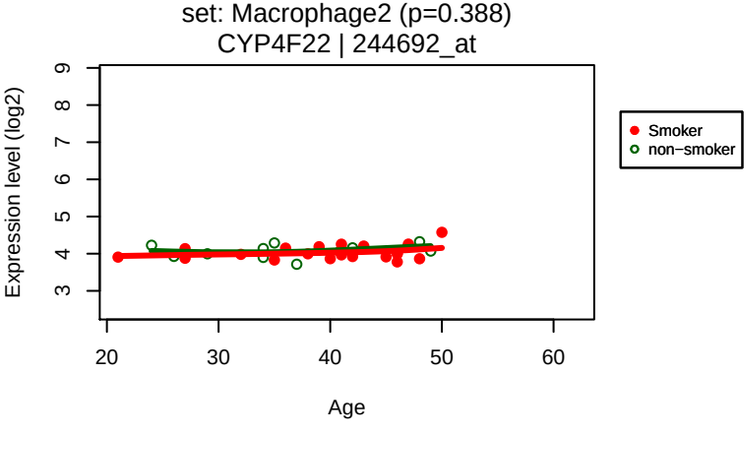

set: SmallAirway2 (p=0.3431)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker

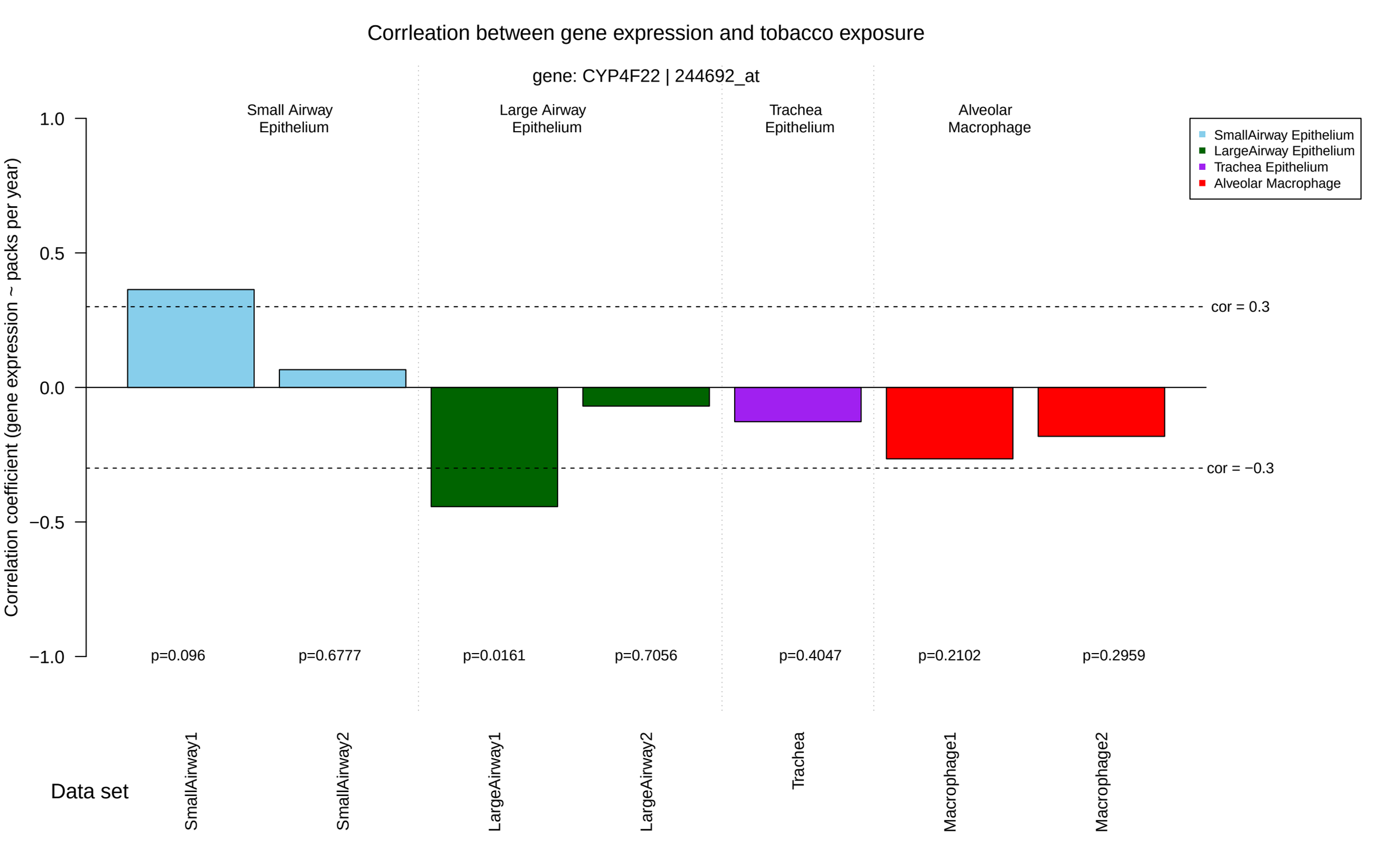

set: LargeAirway1 (p=0.0148)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5498)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6999)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3849)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.388)
CYP4F22 | 244692_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP4F22 | 244692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.096
p=0.6777
p=0.0161
p=0.7056
p=0.4047
p=0.2102
p=0.2959
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2